

Requirements for:

Enterprise Architecture Registry

Friday, December 31, 2010

Requirements Development Team

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Document Change History

The document history shall be cleared and stated as version 1.0 after the requirements have been ratified.

Date	Version No.	Change Description
mm/dd/yyyy	1.0	First Draft

Instructions for Use

Characteristics of a requirement are:

- Each requirement accurately describes the functionality to be built.
- Each requirement should be stated in such a way so that it can be interpreted in only one way.
- Each requirement shall have the means to prove that the system satisfies the requirement.
- Each requirement shall be capable of being independently verified.
- Each requirement shall be written with proper grammar, spelling, punctuation, and in short direct sentences.
- Each requirement shall contain the word “shall.”

Requirements are complete when the following is achieved:

- They are Feasible. It must be possible to implement the requirement within the known capabilities and limitations of the system and its operating environment.
- They are Prioritized. Assign an implementation priority to each requirement.
- They are Comprehensive. No requirement or necessary information is absent.
- They employ the use of consistent terms throughout the document.
- Where applicable, use lists, figures, graphs and tables to present information visually.
- They are uniquely numbered to allow traceability throughout the life of the project.

The blank requirements template is structured so that styles, formatting, and numbering are automatic. To make use of the automated formatting, only use the styles “Heading 1” through “Heading 4”, and “Body” for narrative, non-numbered information.

For tables and individual requirements, unique numbers are automatically generated. Upon ratification, all tables shall be protected so that the unique numbers shall never change.

1 Project Introduction

1.1 Opportunity Description

Develop a consolidated platform for capturing an Enterprise Architect's portfolio of knowledge, skills, and experience which has been validated using a rigorous qualification process to ensure truth, transparency, and promote trust.

1.2 Business Need

The Enterprise Architecture Profession requires a comprehensive set of services to support practitioner and executive career goals:

- Provides guideposts for practitioners who want to achieve professional status
- Promote individual achievement along the road to becoming professional
- Establishes a key structural element which supports the profession of enterprise architecture

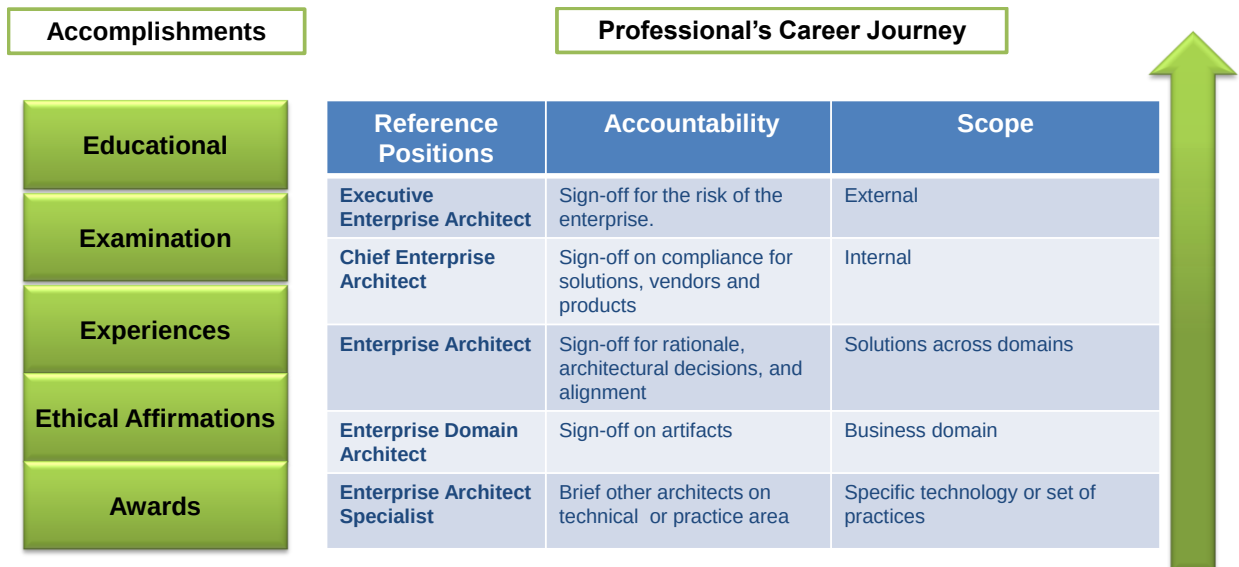
1.3 Justification

No central place to find qualified enterprise architecture professionals

- Enterprise Architecture (EA) Community is splintered
- No consistency amongst Enterprise Architecture practitioners
- Lack of identity and recognition of Enterprise Architecture practitioners
- Unclear value proposition for EAs amongst the public
- Enterprise Architecture Profession needs to be elevated & universally aligned
- Enterprise Architecture services need to be more consistent to engender trust

1.4 Project Objectives

The Registry shall be designed to capture, validate and promote:



1.5 Expected Outcomes

The Registry shall capture and verify the accomplishments for enterprise architects on their pursuit towards achieving a professional status.

Benefits to Registry Members

- *Professional Promotion:*
 - Provides a singular view of your portfolio of experience & achievements
 - Individual achievements publicized through monthly press release and newsletter
- *Career Planning:*
 - A personal dashboard to monitor your career growth
- *Public Visibility:*
 - Trusted source for those who interact or engage enterprise architects
- *Trust and Stability:*
 - Promotes a self-regulated profession

Benefits to the Public & Profession

- *For the Public:*
 - Assure the public they're dealing with a competent enterprise architecture professional
- *For the Profession:*
 - Strengthen the identity
 - Differentiation from other professions
 - Establish consistency and relevance

1.6 In-Scope

A Profile for Your Accomplishments

- | | |
|--|--|
| <ul style="list-style-type: none"> • General Information <ul style="list-style-type: none"> • John Q. Public • Registered Enterprise Architect • Executive Enterprise Architect • Registration <ul style="list-style-type: none"> • Registered Enterprise Architect 2000 • Education <ul style="list-style-type: none"> • Professional Masters in Enterprise Architecture, University of XXX 1990 • Bachelors in Business, Computer Science Minor, University of YYY 1987 • Certifications <ul style="list-style-type: none"> • TOGAF v. 9 2009 • CEnterprise Architecture[DODAF] 2007 • Certified Business Architect CBA 2004 • Continuing Education <ul style="list-style-type: none"> • Seminar in Enterprise Architecture by Zachman 2002 | <ul style="list-style-type: none"> • Experience with the Practice <ul style="list-style-type: none"> • Executive Enterprise Architect, 2007-Current, TTT Company, Banking, Reporting in to SVP of Acquisitions • Chief Enterprise Architect, 2005-2007, DDD Company, Banking, Reporting in to SVP of IT • Chief Enterprise Architect, 2004-2005, FFF Consulting, various engagements • GGG Company, May 2005-July 2005, Health, Reporting in to CEO • ZZZ Company, May 2004-May 2005, Insurance, Reporting in to CIO • Enterprise Architect, 2000-2004, HJK Company, Insurance, Reporting to Chief Enterprise Architect • Service to Profession <ul style="list-style-type: none"> • Volunteer to CAEAP 2009-Current • Contributed to Professional Practice Guide • Member of SIM • Special Interest Group on Enterprise Architecture • Service to the Public <ul style="list-style-type: none"> • External Speaking Engagements • External Trade Publications • External Peer Publications • Case Studies (Portfolio and Professional Style) <ul style="list-style-type: none"> • Large Insurance Company Initiative (link) • Large Bank Acquisition (link) |
|--|--|

A Dashboard for Your Career Path

Education	Examination	Experience	Ethics	Enterprise Architecture Position
●	●	●	●	Executive Enterprise Architect
●	●	●	●	Chief Enterprise Architect
●	●	●	●	Sr. Enterprise Architect
●	●	●	●	Enterprise Architect

Key

- Requirements Fully Met
- Requirements Partially Met
- Requirements Not Met

All Four Greens:

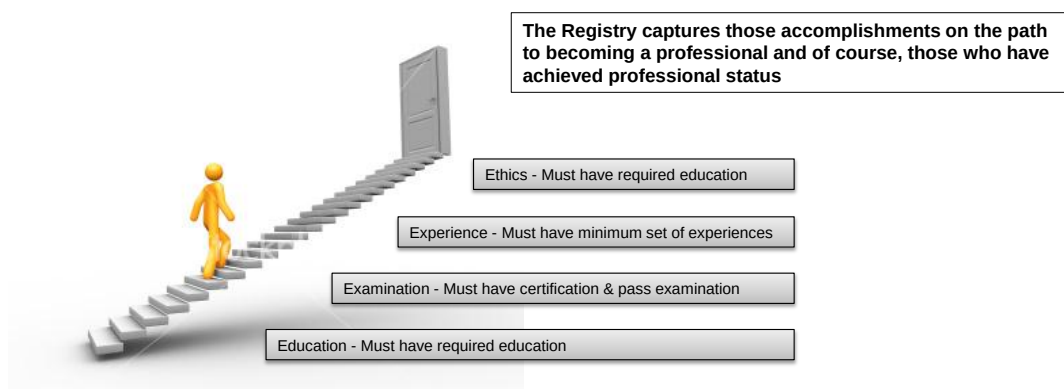
- Call yourself a Registered Enterprise Architect Professional
- Validated to function in a position
- A unique number that identifies you as a professional
- Premier status and recognition in the industry

Maintenance of professional qualifications through service to the profession and continuing education

REA

Constant growth to being a professional enterprise architect through training, experience, mentoring, and education.

1.7 System Context



1.8 Stakeholders and Users



1.9 Risks

Shortage of resources and support from:

- Research: What do practitioners want?
- Recurring cost of the Registry technology platform
- Accepted member qualification process

1.10 Assumptions

Add assumptions here

1.11 Constraints

Add constraints here

1.12 Project Timeline

The Registry captures and verifies the accomplishments for enterprise architects on their pursuit towards achieving a professional status.

Registry Roadmap									
Phase	Category	Phase Deliverables	Phase I			Phase II		Phase III	
1	Establish	Kickoff and Presentation Establishing the Registry requirements Initiate Council of Enterprise Architecture Advisors * Establish Communication Plan Member Signup							
2	Transition	Communication & Promotion Expand Funding Model Initiate Member Qualification Establish Partnerships Begin Service Delivery							
3	Grow	Ramp up Qualification Process Expand Membership (Int'l) Expand Service Delivery Expand Participatory Model Drive Continuous Improvement							

2 Functional Requirements

Functional requirements specify what the system must do. They describe an action the system must take if it is to carry out the work for which it was intended.

“**Owner**” names the person, business unit, or source that generated the requirement and will need to be consulted if the solution cannot meet that particular requirement, or if further clarification is required. The “Owner” field can be used for all requirements or only critical requirements.

“**Critical**” denotes whether a requirement is a “must-have” for the solution to be acceptable, and will require testing in order to ensure that it’s functionally available. Requirements deemed as “critical” are non-negotiable in nature and will direct the development of a solution and/or provide clear selection criteria in the case of a purchased solution.

Below is a temporary notes section:

Profession - Networking Relationships	Manage relationships with other entities / organizations (extra-networking). Those entities may include: • Members who have their own firms • Members who are part of larger organizations • 3rd party organizations • Government Bodies (city, state / region, country) • Education centers • Institutions • Commercial entities (companies) Also need to support the ability of members to capture their own relationships between each other (intra-networking).
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User - Search / Profiling	Need to support the following search & profiling capabilities: • Profiling (type of search, contextual classifications, functionality) • Trending for utilization of search, access to content, functionality • Basic connectors to/from other public services (LinkedIn, Facebook, Twitter, other organizations who also maintain membership registries) • Aggregation of search data for mining
User - References (general)	References include the following types: • Friends • Workers • Employers • Customers • Associates • Organizations
User - References (performance)	360 Feedback should cover the following areas: • Leadership • Technology • Domain knowledge / expertise
Profession - Partnership with other Registries	Hook into other 3 rd party registries
System - Workflow & Escalation	• Verification • Testing • Assessment
System - Payables & Receivables	• Membership Fees • Membership Renewals • Guarantee & Giveback Tracking • Subscription Management • Renewal Processing
Professional - Collateral Repository and Content Push	Need to capture the following artifacts: • Research Materials • Articles • Links & Reference • Surveys & Studies • Lists & Directories • Promote member accomplishments ◦ Internally ◦ Externally to key Partners / newsletters / HR • Consulting opportunities & mentoring ◦ Partnership with key sites • Education for employers • Education for Recruiters • Glossary and Definitions

2.1 System Functionality

	Requirement	Owner	Critical
2.1.1			X

3 Non-Functional Requirements

Non-functional requirements are properties that the system must possess. They can be characteristics that make the system appealing, usable, reliable, or state its availability of use.

3.1 Look & Feel Requirements

	Requirement	Owner	Critical
3.1.1			X

3.2 Usability Requirements

These requirements describe the degree of usability of the system by the intended users. The system's usability is derived from the abilities of the expected users of the system and the complexity of its functionality.

Usability requirements should cover such things as:

- Efficiency. How quickly or accurately the user can use the product
- Error Rates. For some systems it is critical the user commits very few to no errors.
- Technical Level. How computer savvy are the target customers.

Some examples are:

- Input screens shall have drop down lists of possible data values wherever possible to minimize data entry errors.

	Requirement	Owner	Critical
3.2.1			X

3.3 Performance Requirements

Performance requirements can be broken into the following categories:

- Speed and Latency. Specifies the amount of time available to complete specified tasks. These often refer to response times.

- Safety. Quantification of perceived risk of possible damage to people, property and environment.
- Precision. Quantification of the desired accuracy of the results produced by the system.
- Reliability and Availability. This is usually expressed as the allowable time between failures or the total allowable failure rate.
- Robustness. Specifies the ability of the system to continue to function under abnormal circumstances.
- Capacity. Specifies the volumes of data the system must be able to handle. It also quantifies the number of simultaneous users.
- Scalability and Extensibility. Specifies the expected increases in size the system must be able to handle.

Some examples are:

- The system shall have a response time of ≤ 2 seconds between screen refreshes.
- The system shall be able to handle 100 concurrent users.

	Requirement	Owner	Critical
3.3.1			X

3.4 Operational Requirements

Operational requirements can be broken into the following categories:

- Expected Physical Environment. Highlights conditions that might need to be present such as used outside, in field worker's truck, etc.
- Quantifies the expected availability of the system.

Some examples are:

- The system shall be used in a noisy environment with lots of dust.
- The system shall be run on a PDA device.

	Requirement	Owner	Critical
3.4.1			X

3.5 Maintainability and Support Requirements

This section specifies any maintenance and support requirements.

Some examples are:

- The system shall be available 7 days a week, 24 hours a day with the exception of a to be determined maintenance window.
- The system shall have 24-hour phone support.

	Requirement	Owner	Critical
3.5.1			X

3.6 Security Requirements

Security requirements can be broken into the following categories:

- Access. Specifies who is authorized to access the system, under what circumstances access is granted and to what parts of the system access is allowed.
- Integrity. Specifies required integrity of the databases and the system itself.
- Privacy. Specifies what the system must do to ensure the privacy of the individuals it stores information about.
- Audits. Specifies what the system must do to pass required audit checks.
- Immunity. Specifies what the system must do to protect from infection by unauthorized or undesirable software programs such as viruses, worms, etc.

Some examples are:

- System shall use Active Directory Service/LDAP functionality to identify and authenticate users.
- Each user shall be uniquely identified.
- Each user's access permission shall be role-based; i.e., roles shall be defined within the application.
- Access to transactions and access levels shall be defined for each role based on the job needs for the role.
- Users shall be logged out after x minutes that can be set by system administrator.
- All transactions shall log date/time of entry and who created it.
- All activity related to maintenance of application userids, passwords and access permissions shall be logged.

	Requirement	Owner	Critical
3.6.1	The system shall provide access security levels, or sets of permissions, that authorize and define a user's access to the system and their limits of the system's capabilities.	Security	X
3.6.2	The system shall provide the capability to allow a custom logon warning banner (e.g., "Use of the system is for authorized users only") to be displayed during the initial login process.	Security	X
3.6.3	The system shall provide user access via a user name login and password security. A login lockout shall be activated after three (3) unsuccessful login attempts.	Security	X
3.6.4	The system's login interface shall mask, suppress, or otherwise obscure the displayed, copying, or printing of passwords.	Security	X
3.6.5	Upon each unsuccessful login attempt, the system shall not provide any specific information indicating or hinting to the source of the unsuccessful login attempt. A simple message indicating the login was unsuccessful shall be the limit of information displayed.	Security	X
3.6.6	The system shall provide the capability to force general (versus administrator) users to change their passwords every 90 days, and prohibit the general user from re-using a password from the among the user's last 10 passwords.	Security	X
3.6.7	The system shall provide the capability to force administrator users to change their passwords every 30 days, and prohibit the administrator user from re-using a password from the among the administrator user's last 10 passwords.	Security	X
3.6.8	Requests for user accounts to the system shall follow CAEAP established process for requesting user access and Ids.	Security	X
3.6.9	The system shall return to a secure or lock-out state upon idle-timeout (a prescribed period of inactivity). The idle-timeout interval value shall not exceed 15 minutes.	Security	X

3.6.10	The system's capabilities and functions shall not be detrimental to existing communication systems & networks.	Security	X
3.6.11	The system must be able to integrate with security solutions on existing communication systems & networks.	Security	X
3.6.12			

3.7 Business Continuity

These requirements describe what must be provided to the business to continue their operations in the unplanned and unforeseen event that the system becomes unavailable until the time that the system becomes available again. These types of requirements will depend on the nature of the business and the amount of time that the business can function without their system.

An example is:

- The business shall institute manual processes during unplanned or unforeseen events of less than 48 hours.

	Requirement	Owner	Critical
3.7.1			X

3.8 Disaster Recovery

These requirements describe what must be provided to the business to continue their operations in the event of a *disaster* that prohibits access to the system or the inoperability of the system from the time after the *disaster* occurs until the time that business and system operations return to normal.

Specific disaster recovery requirements must be determined through risk assessment and business requirements.

An example is:

- The system shall be analyzed to determine priority / criticality of system in order to determine restoration priority / sequence.

	Requirement	Owner	Critical
3.8.1			X

3.9 Regulatory Requirements

This section specifies requirements that must be satisfy compliance with state, federal, or other regulatory entity affecting business processes and its proposed system.

	The system shall not have the capability to impede, interfere with, or degrade, any existing solution(s) in place.	Security	X
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3.10 Legal Requirements

This section specifies any legal requirements for the system.

	Requirement	Owner	Critical
3.10.1			X

4 Interface Requirements

These requirements specify what and how data is to be transferred or communicated to and from the proposed system.

4.1 Software Interfaces

Describe the connections between this system and other software components (identified by name and version) including databases, operating systems, tools, libraries, installed applications and integrated commercial components. Identify the data that must be shared across the software components.

	Requirement	Owner	Critical
4.1.1			X

4.2 Hardware Interfaces

Describe the characteristics of each interface between the software and hardware components of the system. This description might include supported device types, the data and control interactions between the software and hardware and the communications protocols to be used.

	Requirement	Owner	Critical
4.2.1			X

4.3 Communication Interfaces

Describe the requirements for any communications functions the system will use including email, web browser, and network communication protocols. Define any pertinent message formatting. Specify communications security or encryption issues, data transfers rates and synchronization mechanisms.

	Requirement	Owner	Critical
4.3.1			X

5 Data Requirements

Describe the requirements for any transformation or migration of existing data, or interfaces with systems that may have specific data needs. If known, identify the tables or provide logical data models, which consist of entity relationship diagrams, entity definitions, relationship definitions and attribute definitions. Also describe the expected volumes of data, data growth patterns, data retention and data integrity requirements.

	Requirement	Owner	Critical
5.1.1			X

6 ESB or Cloud Requirements

Provides data transfer functions between two or more systems (or applications). These services include messaging, transaction monitors, object request brokers, file transfer mechanisms, application servers, workflow engines and portals. (These are technical requirements that will be derived as part of the technical design)

	Requirement	Owner	Critical
6.1.1			X

7 Acronyms

8 Glossary

